Balancing Signal and Noise: Context Saturation in Multimodal Large Language Models for Misinformation Detection

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Abstract

Large language models (LLMs) are increasingly applied to misinformation detection, where contextual evidence plays a central role in the model's decision. Although prior work suggests that retrieved context can improve detection, it remains unclear *how much* context is beneficial and at what point additional information becomes harmful, particularly when the retrieved evidence is noisy, irrelevant, or misleading. In this paper we investigate the relationship between context quantity, context quality, and detection performance. We introduce the notion of the **context saturation effect**: adding contextual information improves performance only up to a threshold, beyond which excessive or misleading context ceases to aid reasoning and instead degrades reliability, increasing misclassification. To study this effect systematically we build an open, reproducible evaluation pipeline that unifies four established misinformation datasets (ClaimReview, FakeNewsNet, Fakeddit, and MuMiN) under a single normalised schema, and that supports controlled *context modes* (minimal, full, and misleading) together with deterministic context-budget ablation. We further outline an **adaptive context selection** component intended to prioritise high-quality evidence while suppressing distracting content. We report preliminary results from a pilot study using a deterministic heuristic baseline over 277 samples, which exhibits a pronounced recall-heavy, false-positive-biased behaviour (accuracy $= 0.61$, recall $= 0.98$, precision $= 0.62$) and substantial variance across datasets. These observations motivate, and provide the experimental scaffolding for, a forthcoming larger study across open- and closed-weight LLMs. Our contribution is therefore twofold: a conceptual framing of context saturation for misinformation detection, and a reproducible framework for measuring it.

Keywords: misinformation detection, large language models, retrieval-augmented generation, context-aware reasoning, evidence selection, context saturation, noisy retrieval, robustness.

1 Introduction

We define **context** as any auxiliary evidence retrieved from, or associated with, a piece of content that may influence the reasoning of a large language model, including textual descriptions, metadata, temporal and source signals, social interactions, external evidence, and cross-modal relationships.

Large language models (LLMs) have rapidly become a default tool for misinformation detection, both through direct prompting and through retrieval-augmented generation (RAG) that grounds predictions in external evidence (Lewis et al., 2020; Gao et al., 2024). The promise of these systems rests on a widely held assumption: that providing more relevant context allows the model to reason more accurately about whether a claim is true or false. Multimodal and social-context misinformation, where a claim is accompanied by images, metadata, publication history, and reader reactions, appears to make this assumption even more attractive, since more signals should, in principle, support better-grounded decisions.

Motivation. Human reasoning offers a cautionary counter-example. People also suffer from information overload: a statement may seem plausible when viewed in isolation, yet appropriate background information can reveal it to be misleading, incomplete, or false. At the same time, an excess of irrelevant or conflicting context can confuse rather than clarify. Recent evidence shows that LLMs share this vulnerability. Models struggle to use information placed in the middle of long inputs (Liu et al., 2024), and they can be easily distracted by irrelevant context even on tasks they otherwise solve reliably (Shi et al., 2023). This raises a practical and largely unanswered question for misinforma-

tion detection.

Problem statement. How much context is genuinely beneficial, and when does context become harmful? If detection performance does not increase monotonically with the amount of retrieved evidence, then simply enlarging context windows or retrieving more passages may be counter-productive, especially in adversarial settings where misleading evidence is plausibly present.

Research questions. We organise our investigation around four research questions:

- **RQ1.** How does context *quantity* affect LLM misinformation detection performance?
- **RQ2.** How does context *quality* influence model reliability?
- **RQ3.** Do different LLMs exhibit different saturation thresholds?
- **RQ4.** Can adaptive context selection mitigate context saturation?

Contributions. This paper makes the following contributions. (1) We articulate the **context saturation effect** as a threshold phenomenon in which detection performance rises with informative context and then degrades as noise accumulates. (2) We identify mechanisms, including attention dilution, contradictory evidence, retrieval redundancy, token competition, and reasoning distraction, through which additional context may harm rather than help. (3) We describe a proposed **adaptive context selection** strategy that balances informative signal against noise. (4) We release an open, reproducible evaluation framework that unifies four misinformation datasets under a single schema and supports controlled context-quantity and context-quality manipulation. (5) We present preliminary pilot results that characterise baseline behaviour and motivate the full study.

2 Related Work

2.1 LLM-based misinformation detection

Misinformation detection has a long history in data mining and natural language processing (Shu et al., 2017), and benchmark resources such as FEVER established fact verification as a retrieval-and-reasoning task (Thorne et al., 2018). More recently, LLMs have been applied through zero-shot and few-shot prompting (Brown et al., 2020), chain-of-thought reasoning (Wei et al., 2022), and RAG-based verification (Lewis et al., 2020). A particularly relevant strand examines whether LLMs can detect machine-generated misinformation, finding that such content can be harder to detect than human-written falsehoods (Chen and Shu, 2024). Multimodal misinformation, where text is paired with images or social signals, adds further complexity and motivates reader-centric framings of the problem (Authors, 2024).

2.2 Information overload and distraction

A growing body of work shows that more input is not uniformly helpful. Shi et al. (2023) demonstrate that appending irrelevant sentences degrades LLM reasoning, while related analyses describe cognitive-overload-like failure modes in which distractor evidence amplifies hallucination. These results suggest that the quantity and relevance of context interact in ways that are not captured by accuracy under a single, fixed retrieval budget.

2.3 Context-length studies

Even when context is relevant, its *position* and *length* matter. Liu et al. (2024) document the "lost-in-the-middle" effect, in which models use information at the beginning and end of long inputs far more effectively than information in the middle. This implies that naively concatenating retrieved evidence can dilute the signal that a model actually attends to, a form of context dilution directly relevant to misinformation detection.

2.4 Context-aware retrieval

RAG grounds generation in retrieved passages and has become a standard mechanism for knowledge-intensive tasks (Lewis et al., 2020; Gao et al., 2024). Much of the surrounding literature focuses on improving retrieval quality, evidence selection, ranking, and knowledge grounding. However, these efforts largely aim to retrieve *better* or *more* evidence rather than to characterise the point at which additional evidence stops helping.

2.5 Research gap

Existing work pursues better retrieval and larger context windows, but pays comparatively little attention to the optimal context threshold, to saturation behaviour, and to the interaction between context and injected noise in the specific setting

of misinformation detection. Our work targets this gap directly by manipulating both context quantity and context quality under controlled conditions.

3 The Context Saturation Effect

3.1 More context is not always better

We define the **context saturation effect** as follows: adding contextual information aids an LLM only up to a certain point; beyond that point, additional context no longer improves performance and may reduce reasoning reliability owing to noise, distraction, redundancy, or misleading evidence. Conceptually, detection performance is expected to be non-monotonic in the amount of retrieved context, rising through an *informative zone* and falling within a *noise-dominant zone*.

3.2 Why context saturation happens

We hypothesise five interacting mechanisms.

- **Attention dilution.** As input length grows, the model distributes attention across more tokens, weakening the influence of the most decisive evidence (Liu et al., 2024).
- **Contradictory evidence.** Larger retrieval sets are more likely to contain conflicting claims, forcing the model to arbitrate between sources it cannot reliably rank.
- **Retrieval redundancy.** Repeated near-duplicate passages consume the context budget without adding information, crowding out genuinely useful evidence.
- **Token competition.** Under a fixed budget, noisy passages displace informative ones, so the effective signal-to-noise ratio falls even when total context grows.
- **Reasoning distraction.** Irrelevant but topically adjacent content can derail an otherwise correct reasoning chain (Shi et al., 2023).

These mechanisms predict that misleading context should be disproportionately damaging, and that its effect should be visible in the false-positive and false-negative rates rather than in accuracy alone.

4 Adaptive Context Selection

Motivated by the saturation effect, we propose an **adaptive context selection** component whose objective is to stop retrieval *before* saturation rather than to maximise the quantity of retrieved evidence. The proposed design comprises three elements: (1) **relevance scoring**, which measures the semantic alignment between each candidate evidence item and the claim; (2) **noise detection**, which flags low-quality, contradictory, or potentially misleading evidence; and (3) **dynamic context filtering**, which selects a context budget per claim so as to retain high-signal evidence while suppressing distracting content. We present this component as a design proposal and reserve its empirical evaluation for the full study (RQ4); the framework described in Section 5 already records the artifacts needed to measure where predictions flip as the context budget changes.

Dataset	Domain	Modality
ClaimReview	Fact-check metadata	Text
FakeNewsNet	News articles	Text + image
Fakeddit	Reddit posts	Text + image
MuMiN	Social-media claims	Text + image

Table 1: Datasets unified under the common schema. ClaimReview supplies metadata only; the remaining datasets expose multimodal content.

5 Experimental Framework

5.1 Task

The core task is binary classification: given a claim or content item, and optionally a set of contextual evidence, the model predicts whether the content is *real* or *fake* and produces a short natural-language explanation. Internally the pipeline encodes labels as $0 =$ real and $1 =$ fake; flat exports reverse this convention ($1 =$ real, $0 =$ fake) for readability.

5.2 Datasets

We unify four established datasets into a single canonical corpus with cross-dataset deduplication and consistent normalisation, summarised in Table 1. ClaimReview provides structured fact-check metadata from the Google/Data Commons feed but, importantly, does not include full article bodies. FakeNewsNet draws on PolitiFact and GossipCop (Shu et al., 2020); Fakeddit contributes fine-grained, multimodal Reddit posts (Nakamura et al., 2020); and MuMiN supplies social-context claims drawn from Twitter/X threads (Nielsen and McConville, 2022).

All records are mapped to a unified schema that preserves provenance: the primary text, a binary mapped label, the original source label, optional

secondary context, modality flags, dataset-specific metadata, and an audit trail of cleaning decisions. Labels that cannot be reliably mapped to the binary target are excluded at the loader boundary rather than forced into an uncertain class, preserving dataset integrity. Cross-dataset duplicates are removed using both exact identifier matching and a content fingerprint over title, primary text, and source name.

5.3 Data cleaning

A uniform cleaning pipeline is applied to every dataset: empty or null text is removed, records with fewer than five words are dropped, control and corrupted characters are stripped, and text is normalised to Unicode NFC form. Every modification is logged per record so that retention rates and removed content are fully auditable.

5.4 Context construction

To study context effects we construct three **context modes** for each record:

- **Minimal** – claim text only, testing classification from surface text alone;
- **Full** – claim plus available body, metadata, dates, and source information, testing whether relevant context helps;
- **Misleading** – claim plus deliberately ambiguous or contradictory context, constructed in a label-blind manner to test whether noise degrades reliability.

Orthogonally, the full context can be evaluated at deterministic **context-budget** levels of 1.0, 0.75, 0.5, and 0.25 using sentence-boundary-aware truncation. This enables measurement of the exact point at which predictions flip as context is progressively removed, providing the empirical basis for the saturation curve.

5.5 Models

The primary inference path uses open-weight instruction-tuned models served locally through Hugging Face Transformers, with Qwen2.5-1.5B-Instruct as the default model (Team, 2025); Gemma and Granite are planned additional open-weight targets. The framework also supports an OpenAI-compatible API path, enabling later comparison against closed-weight systems such as GPT-4o, Claude, and Gemini, and a deterministic heuristic baseline used for pipeline validation. Parameter-efficient fine-tuning via LoRA adapters (Hu et al., 2022) is integrated for optional model adaptation without full retraining.

Dataset	**Acc.**	**Prec.**	**Rec.**	F_1
ClaimReview (n=4)	1.000	1.000	1.000	1.000
MuMiN (n=97)	0.969	1.000	0.969	0.984
Fakeddit (n=86)	0.547	0.547	1.000	0.707
FakeNewsNet (n=90)	0.278	0.278	1.000	0.435
Aggregate (n=277)	0.614	0.620	0.983	0.761

Table 2: Preliminary pilot results using the deterministic heuristic baseline. Recall is uniformly high while precision and accuracy vary sharply across datasets.

5.6 Prompting

Prompts follow a single unified template that emits a structured JSON object containing the classification, a confidence score, a short explanation, a list of reasoning signals, and a flag indicating whether external evidence is required. The framework is designed to support vanilla prompting, chain-of-thought prompting (Wei et al., 2022), and RAG-style prompting (Lewis et al., 2020); an optional grounding layer enriches records with cached lookups from the Google Fact Check Tools API.

5.7 Evaluation

For each run we compute accuracy, precision, recall, and the F_1 score, together with the full confusion matrix, and we stratify these metrics by dataset, context mode, and context budget. We additionally record confidence statistics and a sample of explanations annotated with correctness, and we estimate token-based inference cost. Every run writes normalised records, prompts, predictions, evaluation reports, dataset summaries, a cleaning report, and a manifest of all parameters, so that runs are fully reproducible from the command line.

6 Preliminary Results

We report a pilot study intended to validate the pipeline and characterise baseline behaviour, not to draw final conclusions about context saturation. The pilot uses the deterministic heuristic baseline over a balanced sample of approximately 100 records per dataset, yielding 277 retained samples after cleaning. We emphasise that these numbers reflect a keyword-based baseline rather than a neural LLM, and that the full LLM study is forthcoming.

Table 2 reports the per-dataset and aggregate metrics. Three observations stand out. First, performance varies dramatically across datasets, from near-perfect on ClaimReview and MuMiN to poor on FakeNewsNet. Second, recall is uniformly high (0.97–1.00) while precision is low on Fakeddit and FakeNewsNet, indicating a systematic bias toward predicting *fake*. Third, the aggregate confusion matrix (TP $= 170$, TN $= 0$, FP $= 104$, FN $= 3$) confirms that the baseline predicts *fake* for the overwhelming majority of inputs (98.9%).

This false-positive-biased behaviour is precisely the kind of failure mode that context is meant to correct, and that misleading context is hypothesised to worsen. The pilot therefore serves two purposes: it validates that the unified pipeline, cleaning, evaluation, and reporting operate end to end across all four datasets, and it establishes a baseline against which the effect of context quantity and quality can be measured in the full study. A single-sample sanity check with Qwen2.5-1.5B-Instruct under full context produced a correct prediction with confidence 0.8; we report it only as a smoke test and draw no inference from it.

7 Discussion

Implications for RAG. If detection performance is non-monotonic in retrieved context, then retrieval pipelines optimised solely for recall may transport noise into the model and trigger saturation. The framework's context-budget ablation is designed to expose this trade-off directly and to inform per-claim retrieval budgets rather than fixed top-k retrieval.

Implications for misinformation detection. The pilot's false-positive bias illustrates why trustworthiness and robust reasoning, rather than raw accuracy, should guide evaluation. Misleading context is hypothesised to shift the balance of false positives and false negatives, so reporting the full confusion matrix is essential.

Broader implications. The saturation framing is not specific to misinformation. Domains such as medical and legal reasoning, where evidence is abundant, partly contradictory, and consequential, may exhibit similar thresholds, making adaptive context selection broadly relevant.

8 Limitations

This work reports a preliminary study and is subject to several limitations. The pilot results use a deterministic heuristic baseline rather than a neural LLM, so they characterise the pipeline rather than the saturation effect itself. The evaluation is currently English-only. ClaimReview does not provide full article bodies, which constrains the richness of available context for that source, and dataset-specific sampling and class balance affect cross-dataset comparability. Access to closed-weight models is limited, so comparative thresholds across model families (RQ3) remain to be established. Finally, the adaptive context selection component is presented as a design and has not yet been empirically validated.

9 Conclusion

We have argued that, for LLM-based misinformation detection, context helps only up to a point: beyond a threshold, excessive or misleading context degrades reasoning rather than improving it. We framed this as the context saturation effect, identified the mechanisms likely to drive it, proposed an adaptive context selection strategy to counter it, and released a reproducible framework that unifies four datasets and supports controlled manipulation of context quantity and quality. Preliminary pilot results validate the pipeline and reveal a strongly false-positive-biased baseline that motivates the controlled study to come. Our forthcoming work will populate the saturation curves across open- and closed-weight models and evaluate whether adaptive selection improves robustness.

Author Contributions

Jordan Bell Compaan (Research Assistant) designed and implemented the detection agent, inference pipeline, and model training. Michael C. John-Charles (Research Assistant) constructed and curated the unified dataset. Dr. Yi Peiling supervised the research.

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